Vraj Patel

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EDUCATION

University of Massachusetts Dartmouth | *North Dartmouth, MA* *Expected May 2027*

Bachelor of Science in Computer Science | *GPA: 3.53*

Relevant Coursework: Data Structures & Algorithms, Probability, Machine Learning, AI Agents, Software Engineering

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, CUDA, SQL, R

Machine Learning & Robotics: PyTorch, Scikit-learn, Triton, XGBoost, MuJoCo, AprilTags, Reinforcement Learning (SAC), FlashAttention, INT8 Quantization, Paged KV Cache

Data: Pandas, NumPy, Snowflake, Streamlit

Infrastructure: Linux/Unix, AWS (EC2, Lambda, S3, API Gateway), Docker, FastAPI, Firebase, Flutter, Git, Nsight Compute/Systems

EXPERIENCE

Fleet Robotics – ML Engineer Intern | *Boston, MA* *Jun 2026 – Aug 2026*

- Diagnosed that the MuJoCo simulator was not accumulating noise across steps, collapsing predicted drift to ~0° while the real robot drifted ~20° over 15 pivots; rebuilt the noise model against AprilTag ground truth from 20 real robot runs.
- Engineered AWS/Snowflake data pipelines and Streamlit dashboards turning raw robot telemetry into engineering insights.
- Fine-tuned the Soft Actor-Critic (SAC) navigation policy on the corrected simulator, which predicts ~17° drift against the real robot's ~20°, cutting the sim-to-real gap from ~20° to ~3°.
- Presented findings to technical and non-technical stakeholders while contributing within a ~500K-line production codebase.

SMAST, UMass Dartmouth – Gavin Fay Lab – Research Software Engineering Intern | *Dartmouth, MA* *Sept 2025 – Nov 2025*

- Refactored R codebase for NOAA-funded SSEEP project evaluating offshore wind impacts on fishery surveys.
- Implemented parallel processing techniques, achieving **2x speedup** in computation time for spatial simulations.
- Developed and packaged sseephelpers – an R library to streamline common data operations.

SMAST – Data Science Intern | *Dartmouth, MA* *Jun 2025 – Aug 2025*

- Automated analysis of 38 years of estuary data (10K+ records) across 7 sites, **cutting manual effort 70%** via scalable workflows.
- Developed PowerQuery ETL pipelines, **reducing report turnaround from 3 weeks to 10 minutes (95% faster)**.

Blue Tech Externship – Courage Builder Program | *UUV / Autonomous Systems* *Mar 2026*

- Built CMVS – a simulated autonomous UUV for sonar-based mine detection using MATLAB and Simulink; 1st place in the cohort.
- Delivered an internal workshop on AI-assisted development with Claude for the cohort.

PROJECTS

paged-inference-engine – vLLM-Style Paged Inference Server From Scratch | *PyTorch, Triton, Python*

- Designed a mini vLLM-style inference server from scratch (independent build, no reference to vLLM source): block-based paged KV cache with a custom memory manager and a continuous batching scheduler with skip-threshold and lookahead admission.
- Wrote a queue-driven engine loop packing concurrent prefill/decode requests into a single flat tensor per step for GPU-efficient batched execution.
- Rewrote a FlashAttention-2 Triton kernel to index KV through a block table instead of contiguous buffers, then layered per-block INT8 KV cache quantization on top; validated against PyTorch scaled dot-product attention (SDPA) on scrambled non-contiguous block layouts (~0.001 max diff at fp16).
- Built a CUDA-synchronized benchmark harness (time-to-first-token, time-per-output-token, throughput, concurrency sweep N=1–32); continuous batching scaled paged throughput from 95 to 176 tok/s, and profiling traced the remaining gap to a sequential baseline to fixed per-step kernel launch and block-table overhead at this model size.

cpp-gpu-inference – GPU Inference Stack Built From Scratch | *CUDA, Triton, PyTorch, C++*

- Wrote FlashAttention-2 forward and backward in Triton as a custom autograd Function; peak memory matched PyTorch SDPA from 512 to 8K tokens (258 to 288 MB), never materializing the seq×seq attention matrix.
- Implemented LLM.int8()-style outlier-aware INT8 quantization from scratch; perplexity within 0.02% of the unquantized model, 15/15 test translations word-identical, static memory cut 46.4% including the 52K-vocab output projection.
- Wrote and autotuned a custom Triton matmul kernel within 20% of cuBLAS across 512–4096 in true FP32 (TF32 disabled on both sides).
- Engineered CUDA kernels from first principles (memory coalescing, shared-memory tiling, thread coarsening); Nsight Compute confirmed the coalescing fix, with estimated speedup from uncoalesced loads dropping 43% to 3%.

NutriGrove – On-Campus Nutrition Recommendation App | *Flutter, FastAPI, Firebase, Gemini API, Python*

- Built and shipped a full-stack mobile app (Flutter, FastAPI, Firebase) generating personalized meal recommendations from live dining hall menus based on dietary needs; **live on the App Store and Google Play with 100+ users**.
- **Won 2nd place** at the yc0n1c x Code Night PVD **$12,000 AI + Startup Hackathon** (2025) as a two-person team.

Massachusetts Housing Price Predictor & Recommender System | *Python, Scikit-learn, Pandas*

- Built the full pipeline: a Zillow scraper collecting MA state listings (40+ features), plus cleaning, EDA, and feature engineering.
- Trained an ExtraTreesRegressor to **R-squared ≈ 0.854** on price prediction; published the cleaned dataset on Kaggle.
- Built a property recommender and deployed it in an interactive Streamlit app for price estimates and comparable listings.